ECEN 607: Advanced Analog Circuit Tech

Lab 4: Op Amp Design - I

Pre-lab or Lab Report

Name: FirstName LastName
UIN: ******Last Four Digits

Objectives

- To be able to identify the dominant noise sources.
- To study and design a single-ended telescopic op amp.

Prelab

- **In this prelab, 5.5 V NFETs and PFETs will be used.**
- **Do not forget to show relevant design procedures for each design.**
- **The telescopic op amp must be tested in unity gain configuration.**

Design the telescopic amplifier with output dominant pole shown in Fig. 4-1. Identify the dominant noise sources and find the expression for the input referred noise due to the dominant noise sources. Show the hand-calculations for the input referred noise and report the estimated value.
The specifications are set in Table 4-1. Also, add the biasing circuit that generates Vb1 and Vb2.

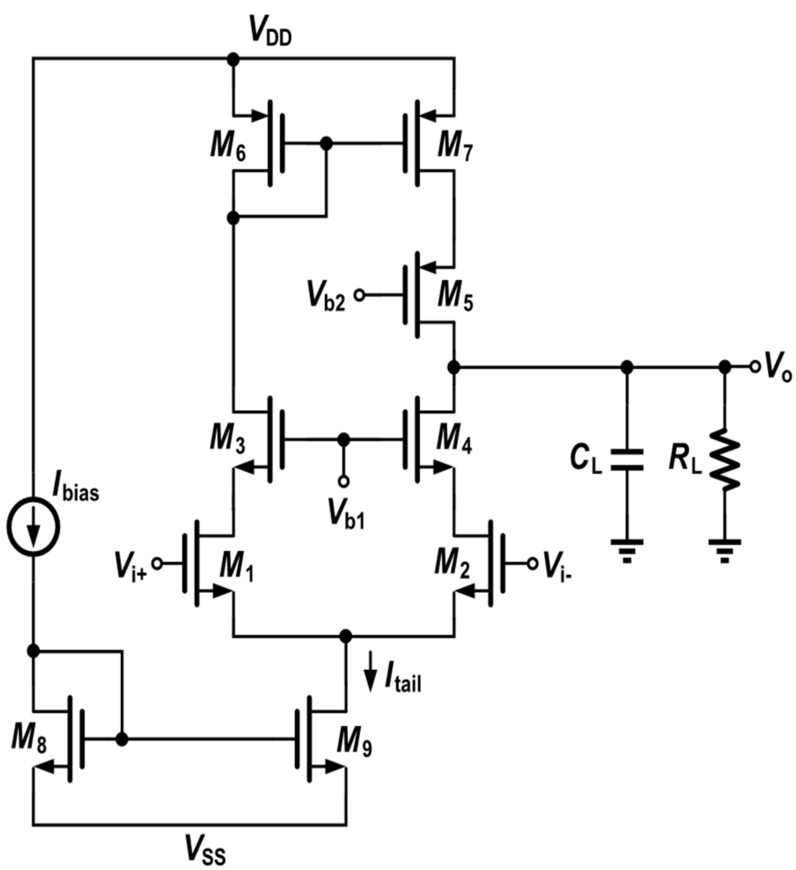

Figure 4-1: Telescopic single stage amplifier

Table 4-1: Design specifications

V_{DD}-V_{SS}	5.5 V
A_{v0}	> 70 dB
GBW	> 2 MHz
PM	> 45°
Output Swing @ gain>40dB	> 0.8 V

C_L	2 pF
R_L	100kΩ
Power Dissipated	< 300 μW
Input referred noise (10 Hz – 2 MHz)	<30 μVrms

Lab Report

Design:

1. Simulate the circuit from the prelab and adjust the transistor sizes accordingly until all specifications are met. Provide necessary plots such that you clearly show the specs are met.

2. Measure the output signal swing; report the proper screen shots.

3. Do a cross simulation (supply variation and temperature variation) and report the worst case DC Gain, GBW, Phase Margin, Gain Margin and input referred noise. Assume there is +/- 5% variation on the supply, -40 <temperature <125 C. You do not have to adjust the design if the specs are not met.

4. Provide comparison tables of hand-calculated vs. final transistor sizes, and required specs vs. simulated specs.

5. Comment on your learning from this lab.